

(MATH203) Applied Linear Algebra

Problems

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Contents

1 (MATH203) Applied Linear Algebra

1

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Problems are arranged chronologically by year and semester, with the midterm preceding the final examination within the same semester, while preserving the original problem order within each source.

Problem 1. (2018 Fall POSTECH Applied Linear Algebra) In this problem, V is a vector space. Determine whether each of the following statements is true or false.

- (1) There exists a linearly independent subset $S \subset V$ that contains the zero vector $\mathbf{0}$.
- (2) If $\mathbf{A} \in \mathbb{R}^{n \times n}$ is invertible, then the columns of $\mathbf{A}$ form a basis for $\mathbb{R}^n$.
- (3) Every projection matrix is invertible.
- (4) $\text{Null}(\mathbf{A})^\perp = \text{Col}(\mathbf{A})$.
- (5) Let $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a basis for V , and let $T : V \rightarrow W$ be a linear transformation. If

$$T(\mathbf{v}_i) = \mathbf{w}_i, \quad 1 \leq i \leq n,$$

for given vectors $\mathbf{w}_1, \dots, \mathbf{w}_n \in W$, then T is uniquely determined.

Problem 2. (2018 Fall POSTECH Applied Linear Algebra) For each statement, prove it or disprove it by providing a counterexample.

- (1) Consider a linear system

$$\mathbf{A}\mathbf{x} = \mathbf{b}, \quad \mathbf{A} \in \mathbb{R}^{m \times n}, \quad \mathbf{x} \in \mathbb{R}^n, \quad \mathbf{b} \in \mathbb{R}^m.$$

If there are more variables than equations, $m < n$, then the system always has a solution.

- (2) If $\mathbf{A} \in \mathbb{R}^{n \times n}$ satisfies

$$\mathbf{A}^2 = \mathbf{A} \quad \text{and} \quad \text{rk}(\mathbf{A}) = n,$$

then $\mathbf{A} = \mathbf{I}_n$.

- (3) The null space of $\mathbf{A}$ is equal to the null space of $\mathbf{A}^\top \mathbf{A}$.

Problem 3. (2018 Fall POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 0 \\ 2 & 1 & 1 \end{pmatrix}.$$

- (a) Find $\mathbf{A}^{-1}$.
- (b) Compute $(\mathbf{A}\mathbf{A}^\top)^{-1}$.

Problem 4. (2018 Fall POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

- (a) Find a right inverse $\mathbf{B}$ such that

$$\mathbf{A}\mathbf{B} = \mathbf{I}_2.$$
- (b) Using the matrix $\mathbf{B}$ obtained in part (a), compute $(\mathbf{B}\mathbf{A})^{100}$.

Problem 5. (2018 Fall POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 4 & 4 \\ 4 & 4 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Find the orthogonal projection $\mathbf{b}_c$ of $\mathbf{b}$ onto $\text{Col}(\mathbf{A})$.

Problem 6. (2018 Fall POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 & 1 & 1 & 1 \\ -1 & 0 & -1 & 0 \\ 1 & -1 & 0 & 0 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 4 & 5 \end{pmatrix}.$$

- Find an LU decomposition $\mathbf{A} = \mathbf{L}\mathbf{U}$, where $\mathbf{L}$ is lower triangular and $\mathbf{U}$ is upper triangular.
- Find a basis for $\text{Col}(\mathbf{B})$.
- Find $\text{Null}(\mathbf{B})$.
- Write the complete solution in the form $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$ for

$$\mathbf{C} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix},$$

where $\mathbf{x}_p$ is a particular solution and $\mathbf{x}_n \in \text{Null}(\mathbf{C})$.

Problem 7. (2018 Fall POSTECH Applied Linear Algebra) Let $V = M_2(\mathbb{R})$, the vector space of 2×2 real matrices with the usual addition and scalar multiplication.

- Show that

$$\mathbf{v}_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{v}_4 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

form a basis for V .

- Find the matrix representation of the linear transformation

$$T : V \rightarrow V, \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \begin{pmatrix} a+d & c \\ b & a-d \end{pmatrix},$$

with respect to the ordered basis $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$.

Problem 8. (2018 Fall POSTECH Applied Linear Algebra) Let

$$P_k = \left\{ a_k t^k + a_{k-1} t^{k-1} + \cdots + a_1 t + a_0 \mid a_0, \dots, a_k \in \mathbb{R} \right\},$$

with the usual polynomial addition and scalar multiplication. Use the ordered basis

$$\{1, t, t^2, \dots, t^k\}$$

for P_k .

- Find the matrix representation $\mathbf{A}$ of the differentiation map

$$D : P_2 \rightarrow P_1, \quad D(f) = \frac{df}{dt}.$$

- Find the matrix representation $\mathbf{B}$ of the integration map

$$T : P_1 \rightarrow P_2, \quad (Tf)(t) = \int_0^t f(s) \, ds.$$

Problem 9. (2018 Fall POSTECH Applied Linear Algebra) Determine whether each of the following statements is true or false.

- (1) $\mathbf{A}$ and $\mathbf{A}^T \mathbf{A}$ have the same rank for every rectangular matrix $\mathbf{A}$.
- (2) The only possible eigenvalues of a symmetric orthogonal matrix are 1 and -1 .
- (3) For an $n \times n$ matrix $\mathbf{A}$ and a scalar $k \in \mathbb{R}$,

$$\det(k\mathbf{A}) = k \det(\mathbf{A}).$$

- (4) For an orthogonal matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ and $\mathbf{x} \in \mathbb{R}^n$,

$$\|\mathbf{Q}\mathbf{x}\| = \|\mathbf{x}\|.$$

- (5) If there is a basis consisting of eigenvectors of a matrix, then the matrix is diagonalizable.
- (6) Every diagonalizable matrix has an orthonormal basis consisting of eigenvectors.

Problem 10. (2018 Fall POSTECH Applied Linear Algebra) Compute the determinants of the following matrices.

- (a) Use cofactor expansion to find the determinant of

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & -1 \\ -1 & 1 & 3 \end{pmatrix}.$$

- (b) Transform the matrix into upper triangular form using elementary row operations, and compute the determinant of

$$\mathbf{B} = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 2 & -1 \\ 2 & 4 & 0 \end{pmatrix}.$$

Problem 11. (2018 Fall POSTECH Applied Linear Algebra) Let $\mathbf{A}$ and $\mathbf{B}$ be real square matrices. Prove that if $\mathbf{A}$ and $\mathbf{B}$ are orthogonally similar, then they have the same singular values.

Problem 12. (2018 Fall POSTECH Applied Linear Algebra) Write down as many pairwise nonsimilar 4×4 matrices as possible whose only eigenvalue is 2.

Problem 13. (2018 Fall POSTECH Applied Linear Algebra) Consider $\mathbb{R}^4$ with the standard inner product and let

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\} \subset \mathbb{R}^4,$$

where

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 2 \\ 3 \\ 2 \\ -3 \end{pmatrix}, \quad \mathbf{v}_4 = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 2 \end{pmatrix}.$$

- (a) Find a subset of S that is a basis for $\text{Span}(S)$ and includes $\{\mathbf{v}_1, \mathbf{v}_2\}$.
- (b) Apply the Gram–Schmidt process to the subset found in part (a) and obtain an orthonormal basis for $\text{Span}(S)$.

Problem 14. (2018 Fall POSTECH Applied Linear Algebra) Identify and sketch the curve

$$x^2 + xy + y^2 = 1.$$

Justify your identification with an explicit equation. Indicate all x -intercepts and y -intercepts, and label all coordinate axes.

Problem 15. (2018 Fall POSTECH Applied Linear Algebra) Solve the initial value problem

$$\mathbf{u}'(t) = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \mathbf{u}(t), \quad \mathbf{u}(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Problem 16. (2018 Fall POSTECH Applied Linear Algebra) Use a similarity transformation to find the matrix $\mathbf{A} \in \mathbb{R}^{3 \times 3}$ representing reflection across the plane

$$x + y + z = 0.$$

Problem 17. (2018 Fall POSTECH Applied Linear Algebra) Find the singular value decomposition of

$$\mathbf{A} = \begin{pmatrix} 4 & 0 & 3 \\ 0 & 0 & 5 \end{pmatrix}.$$

Problem 18. (2019 Spring POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 3 \end{pmatrix}.$$

- (a) Find an orthogonal matrix $\mathbf{P}$ such that $\mathbf{P}^T \mathbf{A} \mathbf{P}$ is diagonal.
- (b) Find the characteristic polynomial of $\mathbf{A}$.
- (c) Show that $\mathbf{A}^{-1}$ can be expressed as

$$a\mathbf{A}^2 + b\mathbf{A} + c\mathbf{I}$$

for some $a, b, c \in \mathbb{R}$, and find a, b, c .

- (d) Find $e^{\mathbf{A}}$.

Problem 19. (2019 Spring POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

- (a) Find all eigenvalues of $\mathbf{A}$.
- (b) Find all eigenvectors of $\mathbf{A}$.
- (c) Determine whether $\mathbf{A}$ is diagonalizable. Explain why.
- (d) Find the Jordan form of $\mathbf{A}$.
- (e) Find a Jordan basis of $\mathbf{A}$.
- (f) Find $\det(e^{\mathbf{A}})$.
- (g) Calculate $e^{\mathbf{A}}$.

(h) Solve

$$\mathbf{y}' = \mathbf{A}\mathbf{y}, \quad \mathbf{y}(0) = \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix}.$$

Problem 20. (2019 Spring POSTECH Applied Linear Algebra) Let

$$\mathbf{D} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{0} & \mathbf{C} \end{pmatrix},$$

where $\mathbf{A}$ and $\mathbf{C}$ are square matrices and $\mathbf{0}$ is a zero matrix of compatible size. Prove that

$$\det(\mathbf{D}) = \det(\mathbf{A}) \det(\mathbf{C}).$$

Problem 21. (2019 Spring POSTECH Applied Linear Algebra) Let (g_k) be a sequence satisfying

$$g_{k+2} = g_{k+1} - g_k, \quad g_1 = 1, \quad g_2 = 2.$$

(a) Find g_{300} .

(b) Determine whether the sequence (g_k) converges. In either case, explain why.

Problem 22. (2019 Spring POSTECH Applied Linear Algebra) Calculate

$$\det \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}.$$

Problem 23. (2019 Spring POSTECH Applied Linear Algebra) Let

$$f(x) = \det \begin{pmatrix} 1 & a & a^2 & a^3 \\ 1 & b & b^2 & b^3 \\ 1 & c & c^2 & c^3 \\ 1 & x & x^2 & x^3 \end{pmatrix}.$$

Find all zeros of $f(x)$ and verify your answer.

Problem 24. (2019 Spring POSTECH Applied Linear Algebra) Let $\mathbf{M} \in \mathbb{R}^{n \times n}$ be invertible, and let $\mathbf{v}_1, \dots, \mathbf{v}_l \in \mathbb{R}^n$ be linearly independent. Prove that

$$\mathbf{M}\mathbf{v}_1, \dots, \mathbf{M}\mathbf{v}_l$$

are also linearly independent.

Problem 25. (2019 Spring POSTECH Applied Linear Algebra) Prove that the two matrices

$$\begin{pmatrix} \mathbf{A}_1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_2 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_3 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} \mathbf{A}_2 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_3 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_1 \end{pmatrix}$$

are similar, where $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3$ are square matrices of sizes m, n, l , respectively.

Problem 26. (2019 Spring **POSTECH** Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & -1+i \\ 0 & 1 & 1-i \\ 0 & 0 & i \end{pmatrix}.$$

- (a) Find the eigenvalues of $\mathbf{A}$.
- (b) Find the eigenvectors of $\mathbf{A}$.
- (c) Find a matrix $\mathbf{S}$ such that $\mathbf{S}^{-1}\mathbf{A}\mathbf{S}$ is diagonal.

Problem 27. (2019 Spring **POSTECH** Applied Linear Algebra) Let $\mathbf{A}$ be a real symmetric matrix, and let λ be an eigenvalue of $\mathbf{A}$ with algebraic multiplicity m . Show that

$$\dim \text{Null}(\mathbf{A} - \lambda\mathbf{I}) = m.$$

Problem 28. (2019 Spring **POSTECH** Applied Linear Algebra)

- (1) State and prove Schur's Lemma.
- (2) State and prove the complex spectral theorem.

Problem 29. (2019 Fall **POSTECH** Applied Linear Algebra) Let $\mathbf{A} = (a_{ij})$ be the $n \times n$ matrix defined by

$$a_{ij} = \begin{cases} 1, & i - j = 1 \text{ and } i < n, \\ 0, & \text{otherwise.} \end{cases}$$

Find the sum of all entries of

$$\sum_{k=1}^{2n} \mathbf{A}^k.$$

Problem 30. (2019 Fall **POSTECH** Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} a & 0 & b \\ a & a & 4 \\ 0 & a & 2 \end{pmatrix}.$$

For what values of a and b does the system

$$\mathbf{A}\mathbf{x} = \begin{pmatrix} 2 \\ 4 \\ b \end{pmatrix}$$

have

- (1) a unique solution,
- (2) a one-parameter family of solutions,
- (3) a two-parameter family of solutions,
- (4) no solution?

Problem 31. (2019 Fall **POSTECH** Applied Linear Algebra) Find the inverse of

$$\mathbf{B} = \begin{pmatrix} 1 & 2 & -3 & 0 \\ 1 & -2 & 0 & 4 \\ -1 & 2 & 0 & 4 \\ 1 & 2 & 3 & 0 \end{pmatrix}.$$

Problem 32. (2019 Fall POSTECH Applied Linear Algebra) Let $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ and $\{\mathbf{f}_1, \dots, \mathbf{f}_n\}$ be two bases of a vector space V . If

$$\mathbf{e}_i = \sum_{j=1}^n a_{ij} \mathbf{f}_j, \quad i = 1, \dots, n,$$

show that the matrix $\mathbf{A} = (a_{ij})$ is invertible.

Problem 33. (2019 Fall POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 5 & 1 & 1 \\ 1 & 1 & 3 & 0 & -1 \\ 2 & 2 & 12 & 1 & 0 \end{pmatrix}.$$

- (a) Find a basis for $\text{Row}(\mathbf{A})$ and a basis for $\text{Col}(\mathbf{A})$.
 (b) Consider the restriction

$$\mathbf{A}|_{\text{Row}(\mathbf{A})} : \text{Row}(\mathbf{A}) \rightarrow \text{Col}(\mathbf{A}).$$

Find its matrix representation with respect to the bases obtained in part (a), and verify that this matrix is invertible.

Problem 34. (2019 Fall POSTECH Applied Linear Algebra) Let $V \subset \mathbb{R}^3$ be a two-dimensional subspace.

- (a) Show that the reflection of any vector $\mathbf{x} \in \mathbb{R}^3$ across V is

$$(2\mathbf{P} - \mathbf{I})\mathbf{x},$$

where $\mathbf{P}$ is the orthogonal projection onto V . *Hint.* Consider an orthonormal basis of V and extend it to an orthonormal basis of $\mathbb{R}^3$.

- (b) Let

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

be the standard basis of $\mathbb{R}^3$. Let V be spanned by

$$\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}.$$

Find the matrix representation of reflection across V with respect to the standard basis.

Problem 35. (2019 Fall POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 2 & 0 & -1 \\ 1 & 2 & -1 & 0 & 2 \\ 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 & 1 \end{pmatrix}.$$

- (a) Find the matrices $\mathbf{P}$, $\mathbf{U}$, and $\mathbf{L}$ in a decomposition

$$\mathbf{PA} = \mathbf{LU}.$$

(b) Solve $\mathbf{Ax} = \mathbf{b}$ for $\mathbf{x}$, where

$$\mathbf{b} = \begin{pmatrix} -2 \\ 3 \\ 5 \\ 4 \\ 1 \end{pmatrix}.$$

Problem 36. (2019 Fall POSTECH Applied Linear Algebra) Find a nonzero matrix $\mathbf{A}$ whose null space is spanned by

$$\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix},$$

and whose left null space is spanned by

$$\begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}.$$

Problem 37. (2019 Fall POSTECH Applied Linear Algebra) Fit a quadratic function

$$f(t) = c_0 + c_1t + c_2t^2$$

to the four data points

$$(-1, 8), \quad (0, 8), \quad (1, 4), \quad (2, 16).$$

Explain why your answer is the best fit.

Problem 38. (2019 Fall POSTECH Applied Linear Algebra) Decompose

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

as

$$\mathbf{A} = \mathbf{QR},$$

where $\mathbf{R}$ is an invertible upper triangular matrix and $\mathbf{Q}$ is an orthogonal matrix.

Problem 39. (2019 Fall POSTECH Applied Linear Algebra) Let P_4 be the vector space of real polynomials of degree at most 4 in the variable t . The map

$$T : P_4 \rightarrow P_4, \quad T(p(t)) = p(t+1),$$

is linear.

(a) Find the matrix $\mathbf{A}$ representing T with respect to the ordered basis

$$\{1, t+1, t^2+1, t^3+1, t^4+1\}.$$

(b) Find $\mathbf{A}^{-1}$.

Problem 40. (2019 Fall POSTECH Applied Linear Algebra) Fit a quadratic function

$$f(t) = c_0 + c_1 t + c_2 t^2$$

to the four data points

$$(-1, 8), \quad (0, 8), \quad (1, 4), \quad (2, 16).$$

Explain why your answer is the best fit.

Problem 41. (2019 Fall POSTECH Applied Linear Algebra) Decompose

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}$$

as

$$\mathbf{A} = \mathbf{Q}\mathbf{R},$$

where $\mathbf{R}$ is an invertible upper triangular matrix and $\mathbf{Q}$ is an orthogonal matrix.

Problem 42. (2019 Fall POSTECH Applied Linear Algebra) For a positive integer n , let S_n be the set of linear combinations of

$$\sin x, \dots, \sin nx, \cos x, \dots, \cos nx,$$

that is,

$$S_n = \left\{ \sum_{k=1}^n (a_k \sin kx + b_k \cos kx) \mid a_k, b_k \in \mathbb{R} \right\}.$$

Then the map

$$D : S_n \rightarrow S_n, \quad D(f) = \frac{df}{dx},$$

is linear.

(a) Find the matrix representation of D with respect to the ordered basis

$$\{\sin x, \cos x, \sin 2x, \cos 2x, \dots, \sin nx, \cos nx\}.$$

(b) Find the determinant and trace of D .

Problem 43. (2019 Fall POSTECH Applied Linear Algebra) Let T be a linear operator on $\mathbb{R}^5$ represented by

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & -2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 3 \\ 1 & 2 & -4 & 0 & 3 \\ 1 & 0 & -1 & 1 & 1 \\ 1 & 2 & -3 & 0 & 2 \end{pmatrix}.$$

(a) Show that the characteristic polynomial has the form

$$p_{\mathbf{A}}(\lambda) = -(\lambda - 1)^3(\lambda - a)^2,$$

and find a .

(b) Find bases for the eigenspaces $E_1(\mathbf{A})$ and $E_a(\mathbf{A})$. Here, $E_1(\mathbf{A})$ (respectively, $E_a(\mathbf{A})$) denotes the subspace of $\mathbb{R}^5$ consisting of all eigenvectors of $\mathbf{A}$ corresponding to the eigenvalue 1 (respectively, a), together with the zero vector.

(c) Find the Jordan form of $\mathbf{A}$.

Problem 44. (2019 Fall POSTECH Applied Linear Algebra) Factorize

$$\mathbf{A} = \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}$$

as

$$\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1},$$

where $\mathbf{U}$ is unitary and $\mathbf{\Lambda}$ is diagonal.

Problem 45. (2019 Fall POSTECH Applied Linear Algebra) Suppose three cities A , B , and C are centers for trucks. Every month, half of the trucks in A and half of those in B go to C , while the other halves stay where they are. Every month, half of the trucks in C go to A and the other half go to B .

(a) Find the Markov transition matrix acting on the initial-state vector

$$\begin{pmatrix} a_0 \\ b_0 \\ c_0 \end{pmatrix},$$

where a_0, b_0, c_0 denote the initial numbers of trucks in cities A, B, C , respectively.

(b) If there are always 450 trucks, find the long-run distribution of trucks.

Problem 46. (2019 Fall POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} a & 1 & 1 \\ 1 & a & 1 \\ 1 & 1 & a \end{pmatrix}.$$

For what range of a is $\mathbf{A}$ positive definite?

Problem 47. (2019 Fall POSTECH Applied Linear Algebra) Find the singular value decomposition of

$$\mathbf{A} = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix}.$$

Problem 48. (2019 Fall POSTECH Applied Linear Algebra) Let $\mathbf{A}$ be an $n \times n$ matrix.

(a) Show that $\mathbf{A} = \mathbf{0}$ if

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (-\lambda)^n$$

and $\mathbf{A}$ is Hermitian.

(b) Show that $\mathbf{A}^n = \mathbf{0}$ if

$$\det(\mathbf{A} - \lambda\mathbf{I}) = (-\lambda)^n.$$

Hint. Consider the Jordan form.

Problem 49. (2019 Fall POSTECH Applied Linear Algebra) Let λ_0 be an eigenvalue of a square matrix $\mathbf{A}$, and let m be the algebraic multiplicity of λ_0 in the characteristic polynomial of $\mathbf{A}$. Show that

$$\dim E_{\lambda_0}(\mathbf{A}) \leq m.$$

Problem 50. (2020 Spring POSTECH Applied Linear Algebra) Determine whether the following statement is true or false: the product of triangular matrices is a triangular matrix.

Problem 51. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{D}$ be an $n \times n$ diagonal matrix. Determine whether the following statement is true or false:

$$\mathbf{AD} = \mathbf{DA}$$

for every $n \times n$ matrix $\mathbf{A}$.

Problem 52. (2020 Spring POSTECH Applied Linear Algebra) Determine whether the following statement is true or false: every nonsingular symmetric square matrix $\mathbf{A}$ admits an LDL^T factorization, where $\mathbf{L}$ is lower triangular and $\mathbf{D}$ is diagonal.

Problem 53. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ be a square matrix. Determine whether the following statement is true or false:

$$\dim \text{Col}(\mathbf{A}^2) = \dim \text{Col}(\mathbf{A}).$$

Problem 54. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ be a rectangular matrix. Determine whether each of the following statements is true or false.

- (1) $\text{Null}(\mathbf{A}^T \mathbf{A}) = \text{Null}(\mathbf{A})$.
- (2) $\text{Null}(\mathbf{A} \mathbf{A}^T) = \text{Null}(\mathbf{A})$.
- (3) If $\mathbf{A}$ is square, then $\text{Null}(\mathbf{A}^2) = \text{Null}(\mathbf{A})$.

Problem 55. (2020 Spring POSTECH Applied Linear Algebra) Determine whether the following statement is true or false: if distinct nonzero vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ span a subspace V of $\mathbb{R}^5$, then there exist mutually orthonormal vectors $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \mathbf{w}_4$ that span V .

Problem 56. (2020 Spring POSTECH Applied Linear Algebra) Let

$$\mathbf{v}_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix},$$

$$\mathbf{v}_4 = \begin{pmatrix} 0 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{v}_5 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{v}_6 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}.$$

Select all true statements.

- (1) $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ span $\mathbb{R}^4$.
- (2) $\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6$ span $\mathbb{R}^4$.
- (3) $\mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6$ are linearly independent over $\mathbb{R}$.
- (4) $\mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5$ are linearly independent over $\mathbb{R}$.
- (5) $\mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5$ span $\mathbb{R}^3$.
- (6) None of the above.

Problem 57. (2020 Spring POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ 1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{3} & 1 & 0 \\ \frac{1}{3} & \frac{1}{4} & 1 \end{pmatrix} \begin{pmatrix} 3 & 1 & -1 \\ 0 & \frac{8}{3} & \frac{4}{3} \\ 0 & 0 & 0 \end{pmatrix}.$$

Select all true statements.

- (1) The vectors

$$\begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 \\ 8 \\ 4 \end{pmatrix}$$

span $\text{Row}(\mathbf{A})$.

- (2) The vectors

$$\begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} -1 \\ \frac{4}{3} \\ 0 \end{pmatrix}$$

span $\text{Col}(\mathbf{A})$.

- (3) The left null space $\text{Null}(\mathbf{A}^T)$ is spanned by

$$\begin{pmatrix} -1 \\ -1 \\ 4 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

- (4) $\mathbf{A}$ has a left inverse.
 (5) $\mathbf{A}$ has a right inverse.
 (6) None of the above.

Problem 58. (2020 Spring POSTECH Applied Linear Algebra) For $a, b, c, d \in \mathbb{R}$, let

$$\mathbf{A} = \begin{pmatrix} a & b & 0 & 0 \\ c & d & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & c & d \end{pmatrix}.$$

Select all true statements.

- (1) $\mathbf{A}$ is invertible if $abcd \neq 0$.
 (2) $\mathbf{A}$ has a right inverse if $cd = 0$.
 (3) $\mathbf{A}$ has both a right inverse and a left inverse if $ad - bc \neq 0$.
 (4) $\mathbf{A}$ has a left inverse if $ab \neq 0$.
 (5) None of the above.

Problem 59. (2020 Spring POSTECH Applied Linear Algebra) Let P_n be the vector space of real polynomials of degree at most n . Let E_n be the set of even polynomials in P_n , that is, polynomials p satisfying $p(x) = p(-x)$, and let O_n be the set of odd polynomials in P_n , that is, polynomials q satisfying $q(x) = -q(-x)$. Determine whether each of the following statements is true or false.

- (1) E_n is a subspace of P_n .
 (2) O_n is a subspace of P_n .

- (3) The map

$$P_{n+1} \rightarrow P_n, \quad f(x) \mapsto \frac{df}{dx},$$

is linear.

- (4) The map

$$P_n \rightarrow P_{2n+2}, \quad f(x) \mapsto \int_1^{x^2} f(u) \, du,$$

is linear.

- (5)
- $\dim P_n = n$
- .

Problem 60. (2020 Spring POSTECH Applied Linear Algebra) Let V be a one-dimensional vector space. Determine whether the following statement is true or false: if $f : V \rightarrow \mathbb{R}$ satisfies

$$f(a\mathbf{v}) = af(\mathbf{v})$$

for every $a \in \mathbb{R}$ and $\mathbf{v} \in V$, then f is linear.

Problem 61. (2020 Spring POSTECH Applied Linear Algebra) Let $V = M_3(\mathbb{R})$, the vector space of all 3×3 real matrices. Determine whether each of the following statements is true or false.

- (1) $\dim V = 9$.
- (2) The subset of invertible 3×3 matrices is a subspace of V .
- (3) The subset of 3×3 matrices whose entries sum to zero is a subspace of V .
- (4) The set of all linear transformations $V \rightarrow V$ is a vector space.
- (5) The matrix representation of a linear transformation $V \rightarrow V$ is a 3×3 matrix.

Problem 62. (2020 Spring POSTECH Applied Linear Algebra) Let V be a vector space and let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$. Determine whether each of the following statements is true or false.

- (1) If $\mathbf{x}, \mathbf{y}, \mathbf{z}$ are linearly independent, then $\mathbf{x}, \mathbf{y}$ are linearly independent.
- (2) If $\mathbf{x}, \mathbf{y}, \mathbf{z}$ are linearly dependent, then

$$\text{Span}(\mathbf{x}, \mathbf{y}, \mathbf{z}) = \text{Span}(\mathbf{x}, \mathbf{y}).$$

- (3) If $\mathbf{x}, \mathbf{y}, \mathbf{z}$ are linearly independent, then

$$\dim \text{Span}(\mathbf{x}, \mathbf{y}, \mathbf{z}) = 3.$$

- (4) If $\dim \text{Span}(\mathbf{x}, \mathbf{y}, \mathbf{z}) = 3$, then $\mathbf{x}, \mathbf{y}, \mathbf{z}$ are linearly independent.
- (5) $\{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ is a basis for $\text{Span}(\mathbf{x}, \mathbf{y}, \mathbf{z})$.

Problem 63. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. Determine whether each of the following statements is true or false.

- (1) $\mathbf{Ax} = \mathbf{b}$ has a solution if and only if $\mathbf{b} \in \text{Col}(\mathbf{A})$.
- (2) $\mathbf{Ax} = \mathbf{b}$ has a unique solution if and only if $\text{rk}(\mathbf{A}) = n$.
- (3) If $\text{rk}(\mathbf{A}) = m$, then $\mathbf{Ax} = \mathbf{b}$ has a solution for every $\mathbf{b} \in \mathbb{R}^m$.
- (4) If $m < n$, then $\mathbf{Ax} = \mathbf{b}$ has infinitely many solutions.
- (5) If $m > n$, then there exists $\mathbf{b} \in \mathbb{R}^m$ for which $\mathbf{Ax} = \mathbf{b}$ has no solution.

Problem 64. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{U}$ be an echelon form of a matrix $\mathbf{A}$. Determine whether each of the following statements is true or false.

- (1) $\text{Row}(\mathbf{A}) = \text{Row}(\mathbf{U})$.
- (2) $\text{Col}(\mathbf{A}) = \text{Col}(\mathbf{U})$.
- (3) $\text{Null}(\mathbf{A}) = \text{Null}(\mathbf{U})$.
- (4) $\text{Row}(\mathbf{A}) = \text{Col}(\mathbf{A})$.
- (5) $\text{rk}(\mathbf{A}) = \text{rk}(\mathbf{U})$.

Problem 65. (2020 Spring POSTECH Applied Linear Algebra) Let $U \subseteq \mathbb{R}^n$ be a subspace. Determine whether each of the following statements is true or false.

- (1) Every $\mathbf{x} \in \mathbb{R}^n$ can be written uniquely as

$$\mathbf{x} = \mathbf{u} + \mathbf{v}, \quad \mathbf{u} \in U, \quad \mathbf{v} \in U^\perp.$$

- (2) If $\mathbf{u} \in U$ and $\mathbf{v} \in U^\perp$, then $\mathbf{u}$ and $\mathbf{v}$ are linearly independent.
- (3) $U \cap U^\perp = \{\mathbf{0}\}$.
- (4) $\dim U = \dim U^\perp$.
- (5) If $\mathbf{A} \in \mathbb{R}^{n \times n}$ satisfies $\text{Col}(\mathbf{A}) = U$, then

$$\mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top$$

is the projection matrix onto U .

Problem 66. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{x}, \mathbf{y}, \mathbf{z}$ be linearly independent vectors in $\mathbb{R}^5$. Applying the Gram–Schmidt process to $\mathbf{x}, \mathbf{y}, \mathbf{z}$ produces orthonormal vectors $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3 \in \mathbb{R}^5$. Let $\mathbf{A}$ be the 5×3 matrix with columns $\mathbf{x}, \mathbf{y}, \mathbf{z}$, and let $\mathbf{Q}$ be the 5×3 matrix with columns $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$. Select all true statements.

- (1) $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_3$.
- (2) $\mathbf{Q} \mathbf{Q}^\top = \mathbf{I}_5$.
- (3) $\mathbf{Q}$ has orthonormal rows.
- (4) $\text{Col}(\mathbf{A}) = \text{Col}(\mathbf{Q})$.
- (5)

$$\mathbf{u}_1 \mathbf{u}_1^\top + \mathbf{u}_2 \mathbf{u}_2^\top + \mathbf{u}_3 \mathbf{u}_3^\top$$

is the projection matrix onto $\text{Col}(\mathbf{Q})$.

- (6) None of the above.

Problem 67. (2020 Spring POSTECH Applied Linear Algebra) Suppose that $\mathbf{Ax} = \mathbf{b}$ is inconsistent, and let $\mathbf{P}$ be the orthogonal projection onto $\text{Col}(\mathbf{A})$. Determine whether each of the following statements is true or false.

- (1) There is always a least-squares solution.
- (2) The least-squares solution is unique.
- (3) Every least-squares solution satisfies

$$\mathbf{A}^\top \mathbf{Ax} = \mathbf{A}^\top \mathbf{b}.$$

- (4) Every solution of

$$\mathbf{Ax} = \mathbf{Pb}$$

is a least-squares solution.

(5)

$$\mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}$$

is a least-squares solution.

Problem 68. (2020 Spring POSTECH Applied Linear Algebra) Let P_2 be the vector space of real polynomials of degree at most 2. Select the matrix representation of the linear transformation

$$T : P_2 \rightarrow \mathbb{R}^2, \quad p(x) \mapsto (p(2), p'(2)),$$

with respect to the ordered bases $\{1, x, x^2\}$ of P_2 and $\{\mathbf{e}_1, \mathbf{e}_2\}$ of $\mathbb{R}^2$.

(1) $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

(2) $\begin{pmatrix} 2 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix}$

(3) $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix}$

(4) $\begin{pmatrix} 1 & 2 & 4 \\ 0 & 1 & 4 \end{pmatrix}$

(5) $\begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 4 & 4 \end{pmatrix}$

Problem 69. (2020 Spring POSTECH Applied Linear Algebra) Let V be a vector space. Determine whether the following statement is true or false: the matrix representation of the identity transformation

$$I_V : V \rightarrow V, \quad \mathbf{v} \mapsto \mathbf{v},$$

with respect to any choice of bases of V is an identity matrix.

Problem 70. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ matrices. Determine whether the following statement is true or false:

$$\det(\mathbf{A} + \mathbf{B}) = \det(\mathbf{A}) + \det(\mathbf{B}).$$

Problem 71. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ be an $n \times n$ matrix. Determine whether the following statement is true or false:

$$\det(2\mathbf{A}) = 2 \det(\mathbf{A}).$$

Problem 72. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ matrices. Determine whether the following statement is true or false:

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B}).$$

Problem 73. (2020 Spring POSTECH Applied Linear Algebra) For

$$\det(\mathbf{A} - \lambda \mathbf{I}_3) = \det \begin{pmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ a_{21} & a_{22} - \lambda & a_{23} \\ a_{31} & a_{32} & a_{33} - \lambda \end{pmatrix},$$

what is the coefficient of λ^2 ?

- (1) 2
- (2) $a_{11} + a_{22} + a_{33}$
- (3) $-a_{11}a_{22}a_{33}$
- (4) $\det(\mathbf{A})$
- (5) None of the above.

Problem 74. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ be an invertible matrix and let $\mathbf{C}$ be its cofactor matrix. Determine whether the following statement is true or false:

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \mathbf{C}^T.$$

Problem 75. (2020 Spring POSTECH Applied Linear Algebra) A parallelepiped in $\mathbb{R}^3$ has edge vectors

$$\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix}.$$

What is its volume?

- (1) $\frac{10}{3}$
- (2) 5
- (3) 7.5
- (4) 10
- (5) None of the above.

Problem 76. (2020 Spring POSTECH Applied Linear Algebra) Determine whether the following statement is true or false: the determinant of a permutation matrix is always 1.

Problem 77. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ be an $n \times n$ matrix and $\mathbf{b} \in \mathbb{R}^n$. Determine whether the following statement is true or false: Cramer's rule gives a solution of $\mathbf{Ax} = \mathbf{b}$ for every such $\mathbf{A}$ and $\mathbf{b}$.

Problem 78. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ be an $n \times n$ matrix. Determine whether the following statement is true or false: if $\mathbf{A}$ has n distinct eigenvalues, then $\mathbf{A}$ is diagonalizable.

Problem 79. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ be an $n \times n$ matrix. Determine whether the following statement is true or false: if $\mathbf{A}$ has fewer than n distinct eigenvalues, then $\mathbf{A}$ is not diagonalizable.

Problem 80. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ be an $n \times n$ matrix. Determine whether the following statement is true or false: eigenvectors of $\mathbf{A}$ associated with distinct eigenvalues are mutually orthogonal.

Problem 81. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ be an $n \times n$ complex matrix. Determine whether the following statement is true or false: $\mathbf{A}$ is unitarily diagonalizable if and only if $\mathbb{C}^n$ has an orthonormal basis consisting of eigenvectors of $\mathbf{A}$.

Problem 82. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ be an $n \times n$ matrix. Determine whether the following statement is true or false: the solution of the initial value problem

$$\mathbf{u}'(t) = \mathbf{A}\mathbf{u}(t), \quad \mathbf{u}(0) = \mathbf{u}_0,$$

is

$$\mathbf{u}(t) = e^{t\mathbf{A}}\mathbf{u}_0,$$

regardless of whether $\mathbf{A}$ is diagonalizable.

Problem 83. (2020 Spring POSTECH Applied Linear Algebra) Determine whether the following statement is true or false: every real symmetric matrix is orthogonally diagonalizable.

Problem 84. (2020 Spring POSTECH Applied Linear Algebra) Determine whether the following statement is true or false: every Hermitian matrix is unitarily diagonalizable.

Problem 85. (2020 Spring POSTECH Applied Linear Algebra) Determine whether the following statement is true or false: every orthogonally diagonalizable matrix is real symmetric.

Problem 86. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ real symmetric matrices. Determine whether the following statement is true or false: if $\mathbf{A}$ and $\mathbf{B}$ are congruent, then they have the same eigenvalues.

Problem 87. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ real symmetric matrices. Determine whether the following statement is true or false: if $\mathbf{A}$ and $\mathbf{B}$ are congruent, then they have the same number of positive eigenvalues.

Problem 88. (2020 Spring POSTECH Applied Linear Algebra) Determine whether the following statement is true or false: every real symmetric matrix is congruent to a diagonal matrix.

Problem 89. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ be a real symmetric matrix. Determine whether the following statement is true or false: if $\mathbf{A}$ is positive definite, then every leading principal submatrix of $\mathbf{A}$ has positive determinant.

Problem 90. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. Determine whether the following statement is true or false: every real matrix has a singular value decomposition.

Problem 91. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. Determine whether the following statement is true or false: the singular values of $\mathbf{A}$ are the nonnegative square roots of the eigenvalues of $\mathbf{A}^T \mathbf{A}$.

Problem 92. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. Determine whether the following statement is true or false: if $\mathbf{A}$ has a left inverse, then its pseudoinverse is a left inverse of $\mathbf{A}$.

Problem 93. (2020 Spring POSTECH Applied Linear Algebra) Determine whether the following statement is true or false: every matrix has a pseudoinverse.

Problem 94. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. Determine whether the following statement is true or false: $\mathbf{A}^\top \mathbf{A}$ and $\mathbf{A} \mathbf{A}^\top$ have the same eigenvalues.

Problem 95. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ be a 2×2 upper triangular matrix with diagonal entries 1 and 2. Determine whether the following statement is true or false: $\mathbf{A}$ is diagonalizable.

Problem 96. (2020 Spring POSTECH Applied Linear Algebra) Determine whether the following matrix is diagonalizable:

$$\mathbf{A} = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}.$$

Problem 97. (2020 Spring POSTECH Applied Linear Algebra) Determine whether the following statement is true or false:

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}^{1000} = \begin{pmatrix} 1 & 0 \\ 0 & 2^{1000} \end{pmatrix}.$$

Problem 98. (2020 Spring POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix}^{-1}.$$

Determine whether the following statement is true or false: the vectors

$$\begin{pmatrix} 3 \\ 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

are eigenvectors of $\mathbf{A}$.

Problem 99. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ be a 2×2 matrix whose eigenvalues are 1 and 2. Determine whether the following statement is true or false: $\mathbf{A}$ is invertible.

Problem 100. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ be a 2×2 matrix whose only eigenvalue is 2. Determine whether the following statement is true or false: $\mathbf{A}$ is not invertible.

Problem 101. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ be a 2×2 matrix. Determine whether the following statement is true or false: if $\mathbf{A}$ is not diagonalizable, then it has only one eigenvalue, counted with algebraic multiplicity two.

Problem 102. (2020 Spring **POSTECH** Applied Linear Algebra) Let $\mathbf{A}$ be a 2×2 matrix with repeated eigenvalue 3. Determine whether the following statement is true or false: $\mathbf{A}$ is invertible.

Problem 103. (2020 Spring **POSTECH** Applied Linear Algebra) Let $\mathbf{A}$ and $\mathbf{B}$ be 2×2 matrices. Determine whether the following statement is true or false:

$$e^{\mathbf{A}+\mathbf{B}} = e^{\mathbf{A}}e^{\mathbf{B}}.$$

Problem 104. (2020 Spring **POSTECH** Applied Linear Algebra) Let $\mathbf{A}$ be an $n \times n$ complex matrix. Determine whether the following statement is true or false: either

$$\det(\mathbf{A}) = \det(\mathbf{A}^H)$$

or

$$\det(\mathbf{A}) = -\det(\mathbf{A}^H).$$

Problem 105. (2020 Spring **POSTECH** Applied Linear Algebra) Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ orthogonal matrices. Determine whether the following statement is true or false: $\mathbf{AB}$ is orthogonal.

Problem 106. (2020 Spring **POSTECH** Applied Linear Algebra) Let $\mathbf{A}$ be a 5×5 matrix with rank 4. Determine whether the following statement is true or false:

$$\det(\mathbf{A}) = 0.$$

Problem 107. (2020 Spring **POSTECH** Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}.$$

Determine whether the following statement is true or false: $\mathbf{A} - \alpha \mathbf{I}_2$ is singular only when $\alpha = 2$ or $\alpha = 5$.

Problem 108. (2020 Spring **POSTECH** Applied Linear Algebra) Determine whether the following statement is true or false: every full-rank 2×2 matrix is diagonalizable.

Problem 109. (2020 Spring **POSTECH** Applied Linear Algebra) Determine whether the following statement is true or false: it is possible to choose $x \in \mathbb{R}$ such that $\det(\mathbf{A}) \neq 0$ for

$$\mathbf{A} = \begin{pmatrix} x & 0 & 0 \\ x & 0 & 0 \\ x & x & x \end{pmatrix}.$$

Problem 110. (2020 Spring **POSTECH** Applied Linear Algebra) Determine whether the following statement is true or false: the only possible eigenvalues of a real symmetric orthogonal matrix are 1 and -1 .

Problem 111. (2020 Spring POSTECH Applied Linear Algebra) Determine whether the following statement is true or false: similar matrices have the same eigenvalues, counting algebraic multiplicity.

Problem 112. (2020 Spring POSTECH Applied Linear Algebra) Determine whether the following statement is true or false: if two $n \times n$ matrices have the same eigenvalues, counting algebraic multiplicity, then they are similar.

Problem 113. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ be a symmetric matrix. Determine whether the following statement is true or false: $\mathbf{A}$ is positive definite if and only if there exists a matrix $\mathbf{R}$ such that

$$\mathbf{A} = \mathbf{R}^T \mathbf{R}.$$

Problem 114. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ be a 4×4 real symmetric matrix with $\det(\mathbf{A}) = 30$. Suppose that $\mathbf{A}$ is not positive definite, all eigenvalues of $\mathbf{A}$ are integers, and two eigenvalues are 2 and 3. What are the other two eigenvalues?

- (1) 1, 5
- (2) -1, -5
- (3) 1, -5
- (4) 2, 3
- (5) None of the above.

Problem 115. (2020 Spring POSTECH Applied Linear Algebra) Determine whether the following statement is true or false: every symmetric matrix with positive determinant is positive definite.

Problem 116. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ be a square matrix. Determine whether the following statement is true or false: if every cofactor of $\mathbf{A}$ is zero, then $\mathbf{A}$ is not invertible.

Problem 117. (2020 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ be a 3×3 matrix with exactly two linearly independent eigenvectors $\mathbf{v}_1, \mathbf{v}_2$, corresponding to eigenvalues λ_1, λ_2 , respectively.

- (a) Determine whether the following statement is true or false: λ_1 and λ_2 must be distinct.
- (b) Select a Jordan form to which $\mathbf{A}$ is similar.

(1) $\begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_2 \end{pmatrix}$

(2) $\begin{pmatrix} \lambda_1 & 1 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_2 \end{pmatrix}$

(3) $\begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 1 \\ 0 & 0 & \lambda_2 \end{pmatrix}$

(4) $\begin{pmatrix} \lambda_1 & 1 & 0 \\ 0 & \lambda_2 & 1 \\ 0 & 0 & \lambda_2 \end{pmatrix}$

- (5) None of the above.

Problem 118. (2020 Spring POSTECH Applied Linear Algebra) Determine whether the following statement is true or false: every complex square matrix has at least one eigenvector.

Problem 119. (2020 Fall POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 4 \\ 2 & 3 & 5 \\ 4 & 5 & 6 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

- (a) Solve $\mathbf{Ax} = \mathbf{b}$.
- (b) Find an LDU decomposition

$$\mathbf{A} = \mathbf{LDU},$$

where $\mathbf{L}$ is lower triangular with diagonal entries 1, $\mathbf{D}$ is diagonal, and $\mathbf{U}$ is upper triangular with diagonal entries 1.

- (c) Find $\mathbf{A}^{-1}$.

Problem 120. (2020 Fall POSTECH Applied Linear Algebra) For

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 4 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{pmatrix},$$

find bases for the row space $\text{Row}(\mathbf{A})$, equivalently $\text{Col}(\mathbf{A}^T)$, and the null space $\text{Null}(\mathbf{A})$.

Problem 121. (2020 Fall POSTECH Applied Linear Algebra) Let V be the subspace of $\mathbb{R}^4$ spanned by the orthonormal vectors

$$\begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}, \quad \begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}, \quad \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix}.$$

Find the orthogonal projection of

$$\mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ -1 \end{pmatrix}$$

onto V .

Problem 122. (2020 Fall POSTECH Applied Linear Algebra) Let V be a subspace of $\mathbb{R}^n$ and let $W = V^\perp$. Show that

$$V + W = \mathbb{R}^n.$$

Problem 123. (2020 Fall POSTECH Applied Linear Algebra) Let $\mathfrak{sl}_2$ be the set of 2×2 real matrices whose diagonal entries sum to zero.

- (a) Show that $\mathfrak{sl}_2$ is a vector space.
- (b) Find a basis for $\mathfrak{sl}_2$, and prove that your chosen matrices form a basis.
- (c) Find $\dim \mathfrak{sl}_2$.

Problem 124. (2020 Fall POSTECH Applied Linear Algebra) Let

$$\mathbf{H} = \mathbf{I} - 2\mathbf{u}\mathbf{u}^\top,$$

where $\mathbf{u}$ is a unit vector.

(a) Show that

$$\mathbf{H}\mathbf{u} = -\mathbf{u}$$

and

$$\mathbf{H}\mathbf{w} = \mathbf{w}$$

for every $\mathbf{w} \perp \mathbf{u}$.

(b) Show that $\|\mathbf{H}\mathbf{x}\| = \|\mathbf{x}\|$ for every $\mathbf{x}$ of compatible dimension.

(c) Let

$$\mathbf{x} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} \sqrt{2} \\ 0 \\ 0 \end{pmatrix}.$$

Find a unit vector $\mathbf{u}$ such that $\mathbf{x}$ solves

$$\mathbf{H}\mathbf{x} = \mathbf{b}.$$

Problem 125. (2020 Fall POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 0 & -1 & -2 \\ -2 & -1 & 2 \\ 3 & -1 & -5 \end{pmatrix}.$$

(a) Find the eigenvalues of $\mathbf{A}$.

(b) Find the eigenvectors of $\mathbf{A}$.

(c) Solve

$$\mathbf{y}'(t) = \mathbf{A}\mathbf{y}(t), \quad \mathbf{y}(0) = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix},$$

and express each component of $\mathbf{y}(t)$ explicitly.

Problem 126. (2020 Fall POSTECH Applied Linear Algebra) Let

$$V = \{\mathbf{v}_1, \dots, \mathbf{v}_n\} \subset \mathbb{R}^n$$

be a set of nonzero mutually orthogonal vectors.

(a) Show that V is a basis for $\mathbb{R}^n$.

(b) Let $\mathbf{A}$ be the $n \times n$ matrix whose columns are $\mathbf{v}_1, \dots, \mathbf{v}_n$. Find $\det(\mathbf{A})$ in terms of $\mathbf{v}_1, \dots, \mathbf{v}_n$.

(c) Let $\mathbf{B}$ be any $n \times n$ matrix. Show that

$$|\det(\mathbf{B})| = \text{vol}(P),$$

where P is the parallelepiped spanned by the columns of $\mathbf{B}$.

Problem 127. (2020 Fall POSTECH Applied Linear Algebra)

- (a) Find a
- 6×6
- matrix
- $\mathbf{A}$
- satisfying

$$\det(\mathbf{A} - \lambda \mathbf{I}_6) = (\lambda - 2)(\lambda - 3)^2(\lambda - 4)^3,$$

and

$$\dim E_2(\mathbf{A}) = 1, \quad \dim E_3(\mathbf{A}) = 1, \quad \dim E_4(\mathbf{A}) = 2.$$

- (b) Compute
- $e^{t\mathbf{A}}$
- for the matrix
- $\mathbf{A}$
- found in part (a).

Problem 128. (2020 Fall POSTECH Applied Linear Algebra) Consider the Markov matrix

$$\mathbf{A} = \begin{pmatrix} 0.5 & 0.5 & 0.25 \\ 0 & 0.5 & 0.25 \\ 0.5 & 0 & 0.5 \end{pmatrix}.$$

- (a) Determine whether the sequence

$$\mathbf{A}^k \begin{pmatrix} 0.25 \\ 0.25 \\ 0.5 \end{pmatrix}$$

converges as $k \rightarrow \infty$.

- (b) Give an example of a Markov matrix
- $\mathbf{M}$
- and a column vector
- $\mathbf{u}_0$
- for which the sequence
- $\mathbf{M}^k \mathbf{u}_0$
- does not converge as
- $k \rightarrow \infty$
- .

Problem 129. (2020 Fall POSTECH Applied Linear Algebra) For

$$f(x, y, z) = 2x^2 + 5y^2 + 8z^2 + 4xy + 6yz,$$

let $\mathbf{A}$ be a symmetric matrix satisfying

$$f(x, y, z) = \mathbf{x}^\top \mathbf{A} \mathbf{x}, \quad \mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

- (a) Find $\mathbf{A}$.
- (b) Show that $\mathbf{A}$ is positive definite using determinants.
- (c) Show that $\mathbf{A}$ is positive definite using pivots.
- (d) Show that $\mathbf{A}$ is positive definite using eigenvalues. *Hint.* 10 is an eigenvalue.

Problem 130. (2020 Fall POSTECH Applied Linear Algebra) Suppose a vector $\mathbf{p}_0 \in \mathbb{R}^3$ rotates at a constant angular rate of 1 rad/s about the unit axis

$$\mathbf{w} = \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix}, \quad \|\mathbf{w}\| = 1.$$

Let $\mathbf{p}(t)$ denote its position at time t . Its velocity is

$$\mathbf{p}'(t) = \mathbf{w} \times \mathbf{p}(t).$$

Since $\mathbf{w} \times \mathbf{p}(t) = \mathbf{\Omega} \mathbf{p}(t)$, where

$$\mathbf{\Omega} = \begin{pmatrix} 0 & -w_3 & w_2 \\ w_3 & 0 & -w_1 \\ -w_2 & w_1 & 0 \end{pmatrix},$$

we have

$$\mathbf{p}'(t) = \mathbf{\Omega} \mathbf{p}(t), \quad \mathbf{p}(0) = \mathbf{p}_0.$$

- (a) Find the eigenvalues of $\mathbf{\Omega}$.
 (b) Show that

$$e^{t\mathbf{\Omega}} = \mathbf{I}_3 + (\sin t)\mathbf{\Omega} + (1 - \cos t)\mathbf{\Omega}^2.$$

Hint. Use the power-series expansions of $\sin t$ and $\cos t$.

- (c) Show that the position vector preserves its length:

$$\|\mathbf{p}(t)\| = \|\mathbf{p}_0\|$$

for all t .

- (d) Suppose

$$\mathbf{w} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{p}_0 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

Find $\mathbf{p}(\pi/2)$.

Problem 131. (2020 Fall **POSTECH** Applied Linear Algebra) Let

$$\mathbf{A} = \mathbf{I}_n - \mathbf{u}\mathbf{v}^\top, \quad \mathbf{u}, \mathbf{v} \in \mathbb{R}^n.$$

- (a) Show that if $\mathbf{A}$ is invertible, then its inverse has the form

$$\mathbf{A}^{-1} = \mathbf{I}_n + \alpha\mathbf{u}\mathbf{v}^\top$$

for some scalar α , and find α .

- (b) Determine for which $\mathbf{u}$ and $\mathbf{v}$ the matrix $\mathbf{A}$ is not invertible.
 (c) If $\mathbf{A}$ is not invertible, find $\text{Null}(\mathbf{A})$.

Problem 132. (2020 Fall **POSTECH** Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

- (a) Find a diagonal matrix $\mathbf{\Lambda}$ and an orthogonal matrix $\mathbf{Q}$ such that

$$\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top.$$

- (b) Compute $\mathbf{A}^{2020}$, describing each entry exactly.
 (c) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation represented by $\mathbf{A}$. Find an orthonormal basis of $\mathbb{R}^3$ with respect to which the matrix representation of T is diagonal.

Problem 133. (2021 Spring **POSTECH** Applied Linear Algebra) Find an LDU decomposition

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 1 & 4 & 4 \end{pmatrix} = \mathbf{L}\mathbf{D}\mathbf{U},$$

where $\mathbf{L}$ is lower triangular with diagonal entries 1, $\mathbf{D}$ is diagonal, and $\mathbf{U}$ is upper triangular with diagonal entries 1.

Problem 134. (2021 Spring POSTECH Applied Linear Algebra) Find an LDU decomposition

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 3 & 1 \\ 2 & 1 & 1 & 4 \end{pmatrix} = \mathbf{L}\mathbf{D}\mathbf{U},$$

where $\mathbf{L}$ is lower triangular with diagonal entries 1, $\mathbf{D}$ is diagonal, and $\mathbf{U}$ is upper triangular with diagonal entries 1.

Problem 135. (2021 Spring POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{pmatrix}.$$

Find a QR decomposition $\mathbf{A} = \mathbf{Q}\mathbf{R}$, where $\mathbf{Q}$ is orthogonal and $\mathbf{R}$ is invertible and upper triangular.

Problem 136. (2021 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ and assume that $\mathbf{A}^\top \mathbf{A}$ is invertible. Define

$$\mathbf{P} = \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top.$$

For $\mathbf{b} \in \mathbb{R}^m$, show that $\mathbf{b} - \mathbf{P}\mathbf{b}$ is orthogonal to every vector in $\text{Col}(\mathbf{A})$.

Problem 137. (2021 Spring POSTECH Applied Linear Algebra) Determine whether each of the following statements is true or false for every matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$. You need not justify your answers.

- (1) $\mathbf{A}$ and $-\mathbf{A}$ have the same reduced row echelon form.
- (2) If $\mathbf{A}$ is invertible, then $\mathbf{A}^\top$ is invertible.
- (3) If $\mathbf{A}\mathbf{x} = \mathbf{b}$ has a solution for every $\mathbf{b} \in \mathbb{R}^m$, then $n \geq m$.
- (4) If $\mathbf{A}\mathbf{x} = \mathbf{b}$ has a unique solution for some $\mathbf{b} \in \mathbb{R}^m$, then $n \leq m$.
- (5) If the columns of a square matrix $\mathbf{A}$ are orthonormal, then its rows are orthonormal.
- (6) If $\mathbf{A}\mathbf{x} = \mathbf{b}$ has infinitely many solutions for some $\mathbf{b} \in \mathbb{R}^m$, then $\mathbf{A}\mathbf{x} = \mathbf{0}$ has a nonzero solution.
- (7) $\text{rk}(\mathbf{A}^\top \mathbf{A}) = \text{rk}(\mathbf{A})$.

Problem 138. (2021 Spring POSTECH Applied Linear Algebra) Find the inverse of

$$\mathbf{B} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ 3 & 4 & 1 & 0 \\ 2 & 3 & 4 & 1 \end{pmatrix}.$$

Problem 139. (2021 Spring POSTECH Applied Linear Algebra) Let P_4 be the vector space of real polynomials of degree at most 4, and define the linear map

$$T : P_4 \rightarrow P_4, \quad T(p(t)) = t^2 \frac{d^2 p(t)}{dt^2}.$$

- (a) Find the matrix $\mathbf{A}$ representing T with respect to the ordered basis

$$\{1, t, t^2, t^3, t^4\}.$$

- (b) Find the matrix $\mathbf{B}$ representing T with respect to the ordered basis

$$\{1, t + 1, t^2 + 1, t^3 + 1, t^4 + 1\}.$$

Problem 140. (2021 Spring POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 4 & 1 & 1 & 1 \\ 1 & 4 & 2 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}.$$

- (a) Find a basis for $\text{Col}(\mathbf{A})$.
 (b) Find a basis for $\text{Col}(\mathbf{A})^\perp$.
 (c) Find an orthonormal basis for $\text{Null}(\mathbf{A})$.

Problem 141. (2021 Spring POSTECH Applied Linear Algebra) State precisely the definition of each of the following concepts.

- (1) Linear independence.
 (2) Orthogonal matrix.
 (3) Vector space.
 (4) Projection matrix.
 (5) Linear transformation.

Problem 142. (2021 Spring POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & -1 & 0 \\ 2 & 4 & 3 \end{pmatrix}.$$

- (a) Find $\mathbf{A}^{-1}$. Write your answer as a single explicit 3×3 matrix.
 (b) Find an LDU decomposition of $\mathbf{A}$.

Problem 143. (2021 Spring POSTECH Applied Linear Algebra) Let

$$V = \text{Span} \left(\begin{pmatrix} 1 \\ 2 \\ 1 \\ 2 \\ -2 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \\ 0 \\ 1 \\ 1 \end{pmatrix} \right) \subset \mathbb{R}^5.$$

- (a) Find $V^\perp$.
 (b) Find all $\mathbf{x} \in \mathbb{R}^5$ satisfying $\mathbf{Ax} = \mathbf{b}$, where

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 & 2 & -2 \\ 2 & 2 & 1 & 1 & 0 \\ 3 & 0 & 0 & 1 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

Problem 144. (2021 Spring POSTECH Applied Linear Algebra) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be reflection across the plane

$$z = 2y.$$

Find the matrix representing T with respect to the standard basis. Write your answer as a single explicit 3×3 matrix.

Problem 145. (2021 Spring POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 2 & 2 & 1 \\ 0 & -1 & -1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

Find the minimum value of

$$\|\mathbf{Ax} - \mathbf{b}\|^2.$$

Problem 146. (2021 Spring POSTECH Applied Linear Algebra) Let V be a vector space. Suppose that

$$\{\mathbf{v}_1, \dots, \mathbf{v}_n\} \quad \text{and} \quad \{\mathbf{w}_1, \dots, \mathbf{w}_m\}$$

are two bases of V . Show that $m = n$.

Problem 147. (2021 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ matrices. Show that if either $\mathbf{A}$ or $\mathbf{B}$ is not invertible, then $\mathbf{AB}$ is not invertible.

Problem 148. (2021 Spring POSTECH Applied Linear Algebra) Let $\mathbf{Q}$ be an $n \times n$ matrix. Show that if

$$\|\mathbf{Qx}\| = \|\mathbf{x}\|$$

for every $\mathbf{x} \in \mathbb{R}^n$, then the rows of $\mathbf{Q}$ are orthonormal.

Problem 149. (2021 Spring POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 2 & 4 & 6 & 4 \\ 2 & 5 & 7 & 6 \\ 2 & 3 & 5 & 2 \end{pmatrix}.$$

- (a) Find an echelon form of $\mathbf{A}$.
- (b) Find $\text{rk}(\mathbf{A})$.
- (c) Find a basis for $\text{Col}(\mathbf{A})$.
- (d) Find a basis for $\text{Col}(\mathbf{A}^T)$.
- (e) Find an LDU factorization of $\mathbf{A}$.
- (f) Solve $\mathbf{Ax} = \mathbf{b}$, where

$$\mathbf{b} = \begin{pmatrix} 16 \\ 20 \\ 12 \end{pmatrix}.$$

Problem 150. (2021 Spring POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 2 & 5 & 7 \\ 3 & 9 & 13 \end{pmatrix}.$$

Show that the columns of $\mathbf{A}$ are linearly independent.

Problem 151. (2021 Spring POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}.$$

Using the Gauss–Jordan method, find $\mathbf{A}^{-1}$.

Problem 152. (2021 Spring POSTECH Applied Linear Algebra) Let $V, W \subset \mathbb{R}^n$ be subspaces.

- (a) Prove that $V + W$ is a subspace of $\mathbb{R}^n$.
- (b) Prove that $V \cap W$ is a subspace of $\mathbb{R}^n$.
- (c) Prove that

$$\dim(V + W) + \dim(V \cap W) = \dim V + \dim W.$$

Problem 153. (2023 Spring POSTECH Applied Linear Algebra) Determine whether each of the following statements is true or false. You do not need to justify your answers.

- (1) Let $\mathbf{A}$ and $\mathbf{B}$ be matrices. If both $\mathbf{AB}$ and $\mathbf{BA}$ are defined, then both products are square matrices.
- (2) Let $\mathbf{A}$ be a square matrix. Suppose that

$$\mathbf{A} = \mathbf{LDU},$$

where $\mathbf{L}$ is lower triangular with all diagonal entries equal to 1, $\mathbf{U}$ is upper triangular with all diagonal entries equal to 1, and $\mathbf{D}$ is diagonal with no zero diagonal entries. Then such a decomposition is unique.

- (3) If $\mathbf{A}$ is a square matrix and $\mathbf{A}^2$ is nonsingular, then $\mathbf{A}$ is nonsingular.
- (4) The set of all $n \times n$ skew-symmetric matrices,

$$\mathbf{A}^\top = -\mathbf{A},$$

forms a subspace of the vector space of $n \times n$ matrices.

- (5) Every subspace of $\mathbb{R}^4$ is the null space of some matrix.
- (6) Let $H_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ denote reflection across the line through the origin making angle θ with the positive x -axis. Then $H_{\theta_1} \circ H_{\theta_2}$ is counterclockwise rotation by $2(\theta_1 - \theta_2)$.
- (7) There exists a nonsquare matrix that has both a left inverse and a right inverse, possibly different from each other.
- (8) Let V and W be finite-dimensional vector spaces. If a linear transformation $T : V \rightarrow W$ is injective, then

$$\dim V \leq \dim W.$$

- (9) Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear. If the values $T(\mathbf{v})$ are known for n distinct nonzero vectors $\mathbf{v} \in \mathbb{R}^n$, then $T(\mathbf{v})$ is determined for every $\mathbf{v} \in \mathbb{R}^n$.
- (10) For a square matrix $\mathbf{A}$, if the columns of $\mathbf{A}$ are linearly independent, then the rows of $\mathbf{A}$ are linearly independent.

Problem 154. (2023 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A} = (a_{ij})_{n \times n}$ be the matrix defined by

$$a_{ij} = \begin{cases} 1, & j - i = 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Evaluate the sum of all entries of $\mathbf{A}$,

$$\sum_{1 \leq i, j \leq n} a_{ij}.$$

- (b) Let $\mathbf{B} = (b_{ij})$ be

$$\mathbf{B} = \sum_{k=0}^{n-1} \mathbf{A}^k = \mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \cdots + \mathbf{A}^{n-1}.$$

Evaluate the sum of all entries of $\mathbf{B}$,

$$\sum_{1 \leq i, j \leq n} b_{ij}.$$

Problem 155. (2023 Spring POSTECH Applied Linear Algebra) Consider

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & 0 \\ 2 & 0 & -2 \\ 0 & a & 4 \end{pmatrix}.$$

- (a) Find all $a \in \mathbb{R}$ for which $\mathbf{A}$ is invertible.
 (b) Using the Gauss–Jordan method, find $\mathbf{A}^{-1}$ when $a = 2$.

Problem 156. (2023 Spring POSTECH Applied Linear Algebra) Consider

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 & 2 \\ 2 & 4 & 8 & 6 \\ -1 & -2 & 1 & 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}.$$

In solving $\mathbf{A}\mathbf{x} = \mathbf{b}$, apply Gaussian elimination to obtain

$$\mathbf{U}\mathbf{x} = \mathbf{c},$$

and then reduce further to

$$\mathbf{R}\mathbf{x} = \mathbf{d},$$

where $\mathbf{U}$ is the echelon matrix obtained from $\mathbf{A}$ and $\mathbf{R}$ is the reduced row echelon form of $\mathbf{A}$. Compute $\mathbf{c}$ and $\mathbf{d}$ in terms of b_1, b_2, b_3 .

Problem 157. (2023 Spring POSTECH Applied Linear Algebra) Let $V = M_2(\mathbb{R})$ be the vector space of 2×2 real matrices, and let P_3 be the vector space of real polynomials of degree at most 3. Consider the linear map $T : P_3 \rightarrow V$ defined by

$$T(a_0 + a_1x + a_2x^2 + a_3x^3) = \begin{pmatrix} a_0 + a_1 & -a_0 - a_2 \\ -a_1 + a_3 & a_2 - a_3 \end{pmatrix}.$$

- (a) Find the matrix $\mathbf{A}$ representing T with respect to the ordered basis

$$\{1, x, x^2, x^3\}$$

of P_3 and the ordered basis

$$\mathbf{E}_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{E}_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{E}_3 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{E}_4 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

of V .

- (b) Find $\text{rk}(\mathbf{A})$ and $\dim \text{Null}(\mathbf{A})$.

Problem 158. (2023 Spring POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 1 & -2 \\ 1 & -1 \\ 1 & 0 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 2 \end{pmatrix}.$$

Decompose

$$\mathbf{b} = \mathbf{b}_c + \mathbf{b}_n,$$

where $\mathbf{b}_c \in \text{Col}(\mathbf{A})$ and $\mathbf{b}_n \in \text{Null}(\mathbf{A}^\top)$, in the following two ways.

- (a) Solve the normal equation

$$\mathbf{A}^\top \mathbf{A} \hat{\mathbf{x}} = \mathbf{A}^\top \mathbf{b}$$

and compute $\mathbf{b}_c = \mathbf{A} \hat{\mathbf{x}}$.

- (b) Find an orthonormal basis $\{\mathbf{q}_1, \mathbf{q}_2\}$ for $\text{Col}(\mathbf{A})$ and express $\mathbf{b}_c$ as a linear combination of $\mathbf{q}_1$ and $\mathbf{q}_2$.

Problem 159. (2023 Spring POSTECH Applied Linear Algebra) Suppose that data are expected to conform to the quadratic relation

$$y(t) = A + Bt + Ct^2.$$

The observed data are

$$\begin{array}{c|ccccc} t & -2 & -1 & 0 & 1 & 2 \\ \hline y(t) & -4 & 1 & 3 & 4 & 6 \end{array}.$$

Find the parameters A, B, C that minimize the sum of squared errors.

Problem 160. (2023 Spring POSTECH Applied Linear Algebra) Let V be a subspace of $\mathbb{R}^d$. Prove that if

$$\{\mathbf{v}_1, \dots, \mathbf{v}_m\} \quad \text{and} \quad \{\mathbf{w}_1, \dots, \mathbf{w}_n\}$$

are both bases for V , then $m = n$.

Problem 161. (2023 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ be an $m \times n$ real matrix. Prove that

$$\text{rk}(\mathbf{A}) = \text{rk}(\mathbf{A}^\top \mathbf{A}) = \text{rk}(\mathbf{A} \mathbf{A}^\top).$$

Problem 162. (2023 Spring POSTECH Applied Linear Algebra) Determine whether each of the following statements is true or false. You do not need to justify your answers.

- (1) There exist 3×3 matrices $\mathbf{A}$ and $\mathbf{B}$ such that

$$\mathbf{AB} - \mathbf{BA} = \mathbf{I}.$$

- (2) If $\mathbf{A}$ and $\mathbf{B}$ are simultaneously diagonalizable, and $\mathbf{B}$ and $\mathbf{C}$ are simultaneously diagonalizable, then $\mathbf{A}$ and $\mathbf{C}$ are simultaneously diagonalizable.

- (3) There exists a real 2×2 matrix $\mathbf{A} \neq \mathbf{I}$ such that

$$\mathbf{A}^3 = \mathbf{I}.$$

- (4) If a 3×3 matrix $\mathbf{A}$ has eigenvalues 1, 2, 3, then

$$(\mathbf{A} - \mathbf{I})(\mathbf{A} - 2\mathbf{I})(\mathbf{A} - 3\mathbf{I})$$

cannot have a nonzero eigenvalue.

- (5) If a real matrix $\mathbf{A}$ is normal, then

$$(\mathbf{A}^k)^+ = (\mathbf{A}^+)^k$$

for every positive integer k .

- (6) If $\det(\mathbf{A}) > 0$ and $\det(\mathbf{B}) > 0$, then

$$\det(\mathbf{A} + \mathbf{B}) > 0.$$

- (7) If $\mathbf{A} = (a_{ij})$ is an $n \times n$ matrix satisfying

$$\sum_{i=1}^n a_{ij} = 5$$

for every j , then 5 is an eigenvalue of $\mathbf{A}$.

- (8) If $\mathbf{A}$ is an $n \times n$ symmetric matrix, then $e^{\mathbf{A}}$ is positive definite.
 (9) Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times n$ matrices. If λ is an eigenvalue of $\mathbf{A}$ and μ is an eigenvalue of $\mathbf{B}$, then $\lambda + \mu$ is an eigenvalue of $\mathbf{A} + \mathbf{B}$.
 (10) If a real symmetric matrix $\mathbf{A}$ is positive definite, then it is nonsingular.

Problem 163. (2023 Spring POSTECH Applied Linear Algebra)

- (a) Find the volume of the parallelepiped determined by the edge vectors from $(0, 0, 0)$ to

$$(-1, 2, 2), \quad (2, -1, 2), \quad (2, 2, -1).$$

- (b) Compute

$$\det \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

Problem 164. (2023 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}, \mathbf{B}, \mathbf{C}$ be $n \times n$ matrices and let $\mathbf{0}$ be the $n \times n$ zero matrix. Show that

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{0} & \mathbf{C} \end{pmatrix} = \det(\mathbf{A}) \det(\mathbf{C}).$$

Hint. One possible approach is to use the Leibniz formula for the determinant.

Problem 165. (2023 Spring POSTECH Applied Linear Algebra) Let $(a_n)_{n=0}^{\infty}$ and $(b_n)_{n=0}^{\infty}$ be sequences satisfying

$$a_{n+1} = 3a_n + 2b_n, \quad b_{n+1} = a_n + 4b_n, \quad a_0 = 1, \quad b_0 = 0.$$

Evaluate

$$\lim_{n \rightarrow \infty} \frac{a_n - b_n}{a_n + b_n}.$$

Problem 166. (2023 Spring POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & 1 & 1 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Find two matrices among $\mathbf{A}, \mathbf{B}, \mathbf{C}$ that are similar to each other. Justify your answer.

Problem 167. (2023 Spring POSTECH Applied Linear Algebra) Let

$$\mathbf{A} = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix}.$$

- (a) Find the singular value decomposition of $\mathbf{A}$.
- (b) Find the pseudoinverse $\mathbf{A}^+$ of $\mathbf{A}$.

Problem 168. (2023 Spring POSTECH Applied Linear Algebra)

- (a) Find all $\alpha \in \mathbb{R}$ such that

$$\mathbf{A} = \begin{pmatrix} 4 & \alpha \\ \alpha & 5 \end{pmatrix}$$

is positive definite.

- (b) For a real number α such that $\mathbf{A}$ is positive definite, let $d_1(\alpha)$ and $d_2(\alpha)$ be the distances from the origin to, respectively, the closest and farthest points on

$$4x^2 + 2\alpha xy + 5y^2 = 1.$$

Find the minimum value of

$$f(\alpha) = d_1(\alpha)^2 + d_2(\alpha)^2.$$

Problem 169. (2023 Spring POSTECH Applied Linear Algebra) Let $\mathbf{A}$ be a square matrix with real entries. Show that

$$\operatorname{tr}(\mathbf{A}^\top \mathbf{A}) = 0$$

if and only if $\mathbf{A} = \mathbf{0}$.

Problem 170. (2023 Spring POSTECH Applied Linear Algebra) For an $n \times n$ matrix $\mathbf{A}$, show that

$$\det(e^{\mathbf{A}}) = e^{\operatorname{tr}(\mathbf{A})}.$$

Hint. By Schur's triangularization theorem over $\mathbb{C}$, there exists an invertible matrix $\mathbf{M}$ such that $\mathbf{M}^{-1}\mathbf{A}\mathbf{M}$ is upper triangular.